

MAIIC_After_Build_Report_2026-10-08

MAIIC EIR and IFRS 9 system: after-build report, 8 October 2026

What this is. On 8 October 2026 the build set out in *MAIIC EIR Engine Specification v4* (7 October 2026) was carried out against the specification, section by section, on the MAICC-IFRS9 repository (branch `eir_revenue_recognition`, commits 32e7d99 to 09ac5da). This report says what was built, what the system proved when it ran, what the build found in the data and in the system that the specification did not know, and what is still owed. It is the comparison the specification asked for in its own section 8 (the build plan): the original specification stands as written; this report is read beside it.

The production database `maiic_ifrs9` was not opened. The build was proven twice: on the demo copy (`maiic_ifrs9_demo`, which also holds the data of the 8 October demo), and on a throwaway database (`maiic_ifrs9_bootstrap`) wiped and rebuilt from nothing by the bootstrap command, which is the test the specification set for the data foundation.

1. In plain language

The specification said the system should load itself from E-Banker's own tables through one door with gates, build the monthly loan book from the ledger, hold the take-on schedules as data, run its engines in order, and prove the result against golden numbers, all from one command on a clean database. That is now what happens: `php artisan eir:bootstrap --fresh --with-client-inputs --build --run-engines --verify` takes a wiped database to a proven state in about 80 seconds, and every golden check passes. The ledger-derived loan book reproduces E-Banker's own carrying amount to the cent on 2,361 of 2,361 account-months. The interest posted from January 2025 to July 2026 is the 5,293,988,207.06 the audit found. The loan GLs at 31 December 2025 tie to the trial balance to the cent, with the two FInES GLs off by exactly the keyed-opening exception Finance has been asked to correct.

The screens moved to the suite layout with light and dark mode, and three screens the specification asked for exist: the E-Banker Feed, the Take-on Schedules and the EIR as at a Date. The transmission methods for the forward-looking adjustment are built as governed choices, each explained in the system with its preconditions checked against MAIIC's data.

Three things the specification did not know were found. The application already holds a **November 2025 Mega Farm loan book** of 3,490 loans across six GLs (K54.85 billion), loaded on 21 January 2026 with no ECL computed; it bears directly on the Mega Farm questions of specification section 16. The production database carries **three columns no migration created**, which a clean install would have lacked; a guarded migration now adds them. And the take-on schedules, compared with E-Banker's opening postings, differ on 72 of 73 mapped accounts, because they are contractual, not actual; the build keeps every take-on loan at its take-on balance, as the governed basis says, until the fees arrive.

What remains is set out in section 6: the forward-looking regression chain and the scenario-set engine (the specification's FL-1 to FL-4 and SC-1 to SC-4), the audit workbooks (CA-1 to CA-4), the Mega Farm module (section 16), the page-by-page dark-mode sweep, and the decisions MAIIC owes.

2. What was built, against the specification

Spec section	What the specification asked for	Built	Where
4.2 Governance Centre	Every open choice as a governed setting with a seeded default	Yes: 48 settings	GovernanceService::catalogue(), GovernanceServiceTest
3.6 / D31 Staging	The directive's thresholds by tenor class; DPD from the oldest overdue instalment; the missed-instalment trigger	Yes	StagingThresholdSeeder, StagingService, eir:stage
6.3 Landing zone	Raw tables mirroring E-Banker, append-only, versioned, every row carrying its pack	Yes, as one physical table with the query id as the logical table	migration create_ebanker_landing_zone, PackLandingService
6.4 Gates	Hashes, counts, dates, register, keys; a failure names the file and row; quarantine	Yes, with key uniqueness and composite keys added	PackLandingService, 10 tests
6.5 Routes	Five routes, the one in force governed	Route 1 built (command and screen upload); routes 2 to 5 are the same door with a different carrier and are not yet wired	eir:land-pack, EirFeedController::land
6.2 / 6.6 Build	Methods A, B and C; maker-checker; differences first; locked periods never restated; ECL columns untouched	Yes	LoanBookBuildService, eir:build-loan-books, eir:lock-period, 11 tests
6.7 History	Built on a copy by A then B and compared	Yes: B equals the stored report to the cent on every account-month but one (1 cent)	the bootstrap's verify
6.8 Feed screen	Queries, loads, watermarks, quarantine, build with approval	Yes	EirFeedController, Pages/Eir/Feed/Index.vue
6.9 Take-on	Landed not typed; blocks and lines with cells; gates; the build under takeon_history_basis; the screen with confirm and fees	Yes	TakeonLandingService, eir:land-takeon, EirTakeonController, 3 tests
6.10 Bootstrap	One command: fresh, seed, land, build, engines, verify	Yes	eir:bootstrap
6.10.1 Engine chain	Twelve engines in dependency order	Steps 1 (macro), 2 (scenario set), 3 (staging), 4 (PD transition matrix), 5 (LGD cohort workout), 6 (forward-looking chain), 7 (ECL), 8 and 9 (EIR), 10 (reconciliation) run; 11 (stress) and 12 (reports and workbooks) remain the screens'	Bootstrap::stepEngines, PdEngineService, LgdEngineService
6.10.2 Golden numbers	The section 9 baselines and the year-end ties	Nine checks, all PASS	Bootstrap::stepVerify
6.11 As at any date	The EIR computation for a loan or the book at any date; refused after the last posting	Yes	EirAsAtService, eir:as-at, EirAsAtController, 4 tests
7.1 Ledger importer	The ledger as the primary record: interest posted and cash movements from it	Yes: interest per account-month (retiring Extract C) and the cash movements as actual transactions (retiring Extract B)	ContractInputsBuildService
7.2 to 7.4	Rate history, contract master, fees from E-Banker's tables	Yes, through the importers that already exist	ContractInputsBuildService
11 Suite layout	Six groups with accents, rail, top bar, page header with breadcrumb, light and dark	Shell, shared styles and the tree built; the page-by-page dark sweep (11.9) remains	config/menu.php, AppLayout.vue, useTheme.js, NavigationTest

Spec section	What the specification asked for	Built	Where
13 Macro statistics	Codes on the series, World Bank fetcher, batches with provenance, the command with a snapshot	MS-1 to MS-3 built; the IMF WEO parser, the RBM file tab and the five-tab screen remain	MacroSeriesSeeder, WorldBankFetcherService, MacroImportService, macro:import-worldbank
14.7 Transmission methods	Governed methods, each explained with live preconditions and a worked example	Yes	TransmissionMethodCatalogue, fli:method-cards, 4 tests
14.3 to 14.6 Forward-looking chain	Guardrail, structural events, proxy deriver, profiler, correlation finder, regression	Yes: the suite's engines ported as they are over a bridge from MAIIC's tables; fli:correlate	FliBridgeService, app/Services/Fli/*, 1 test
15 Scenario sets	The governed set: rules, shocks on the base, approval, lock, versions, back-test, sensitivity; the first set of 15.8	Yes	ScenarioSetService, scenario:sets, Governance Centre / Scenario Sets, 3 tests
12 Audit workbooks	The engine with the two columns and the Baselines sheet; the five modules	Engine ported (CA-1); IFRS 9 EIR and impairment modules drafted and built in three formats (CA-2, part of CA-4); the register, trace and pack (CA-3) and three modules remain	tools/compliance/, docs/compliance/
16 Mega Farm	The programme in the ECL module under D30	Not started; the November 2025 book found (section 4)	section 6

3. What the system proved when it ran

The bootstrap on the wiped database, 8 October 2026 evening, then the engine chain as completed in the second round (about 90 seconds in all):

Step	Result
Land the pack	54 files, 86,000 rows, every gate passed; the five accepted exceptions recorded on the load
Land the take-on workbook	100 blocks, 3,741 lines; 96 mapped, 3 not matched, 1 refused (blocks 34 and 55 both claim account 000104450000036); 64 block principals equal the take-on posting exactly; 96 blocks without a fee row, flagged, never assumed fee-free
Build the loan books	26 periods, July 2024 to August 2026, by method B: 2,749 rows, 21 flagged (one account one cent off the stored report in every reported month)
Take-on population	78 accounts E-Banker migrated; all 78 start at the take-on balance because no fees have been supplied
Contract master	144 created, 35 held (closed before the stored report begins)
PLR series, fees, interest, cash	48 PLR rows with 26 changes; 87 fee lines (arrangement 206.3 m, legal 136.7 m, other 9.5 m); 2,182 account-months of interest posted; the cash movements as actual transactions
Schedules and EIR	31 schedules generated and approved under the bootstrap label; 31 EIRs solved; 12 locked; 19 blocked at the fee gate because the rulebook is unapproved
Revenue	138 roll-forward rows, 12 contracts, January 2025 to August 2026; 107 months take their cash from the ledger
Staging	26 periods; 1,267 / 454 / 1,028 row-months in Stages 1, 2, 3
ECL	August 2026 computed on the pre-FLI PD
PD	Transition matrix over August 2025 to August 2026 (108 transitions): PD to Stage 3 of 35.9% from Stage 1 and 35.2% from Stage 2
LGD	Cohort of 41 Stage 3 loans (4.12bn) followed twelve months: cure 21.2%, recovery 23.5%, LGD 60.25%

Step	Result
ECL, August 2026	8.33bn on 23.32bn gross: Stage 1 1.58bn, Stage 2 1.82bn, Stage 3 4.93bn
Scenario set	The first set of 15.8 approved under the bootstrap label; sensitivity on 135 loans: base 9.63bn, upside 9.00bn, downside 11.10bn, severe 12.99bn, weighted 10.24bn
Forward-looking chain	1,045 driver x proxy x lag evaluations; 209 suggestions (55 recommended); 209 fits, 39 applied by the guardrail, 170 declined with reasons
Verify	9 checks, 0 FAIL (the table in section 3.1)

3.1 The golden numbers

Check	Expected	Actual	Result
Interest posted Jan 2025 to Jul 2026 (ledger, types 303 and 120)	5,293,988,207.06	5,293,988,207.06	PASS
Derived carrying amount against the stored report (over 2 cents)	0 differences	0 of 2,361 account-months	PASS
Year-end interest batch contra on 4215	32,956,675.55	32,956,675.55	PASS
Year-end interest batch contra on 4216	-21,733,262.70	-21,733,262.70	PASS
Loan book 1050101 at 31 Dec 2025 against the trial balance	1,368,737,810.68	1,368,737,810.68	PASS
Loan book 1050102	6,434,045,052.56	6,434,045,052.56	PASS
Loan book 1050201 (accepted exception, spec 3.5: -400,000.00)	1,402,393,139.46	1,401,993,139.46	PASS
Loan book 1050202 (accepted exception, spec 3.5: +1,000,000.00)	3,808,288,656.15	3,809,288,656.15	PASS
Loan book 1050401	1,827,148,120.69	1,827,148,120.69	PASS

The ECL at 31 December 2025 and 2024 against the audited allowance, and the revenue shift, are not yet golden numbers: the first needs the Mega Farm book and the approved PD route; the last is born on the first run Dr Thom approves (spec 6.10.2).

3.2 Staging under the directive, December 2025

Under the governed rule (Stage 2 from 31 days; Stage 3 from 91 days for a facility of 12 months or less and from 181 days otherwise; four consecutive missed instalments) the December 2025 book of 123 accounts stages 57 / 17 / 49. The November 2025 ECL model, on the uniform 181-day rule, staged its 100 loans with fewer in Stage 3. The difference is the directive's short-term rule and the missed-instalment trigger; both are governed and can be set back for comparison.

4. What the build found

The November 2025 Mega Farm loan book. The demo copy of the production database holds, for reporting period 2025-11, 3,490 loans across the Mega Farm GLs: 1050301 fertiliser (1,523 loans, K38.91 bn), 1050302 seed (1,180, K8.61 bn), 1050303 working capital (102, K0.90 bn), 1050304 equipment (130, K2.07 bn), 1050305 irrigation (3, K2.16 bn), 1050307 pesticides (552, K2.20 bn): K54.85 bn in all, loaded on 21 January 2026 at a uniform 15 percent rate with the arrears buckets filled and no ECL computed. Under the MEGA_FARM staging class (91 days) they stage 3,161 / 18 / 436. This is the monthly Mega Farm loan book the email to Dr Thom asks for, at least for one month; it answers part of specification section 16.8 (the book exists in the system) and sharpens the question of how the K39.76 bn provision was arrived at, since the book as loaded is mostly Stage 1 by days past due.

Schema drift. A clean migrate lacked `loan_books.ead`, `off_balance_sheet_exposure` and `customer_name_imported`, which the production database carries and the ECL service and reports read. They were added by hand at some point. Migration `2026_10_08_300000` adds them, guarded, so both a clean install and the production copy end in the same shape. The production copy also carries eleven working tables (`legacy_loanbook`, `stageing_*`, `ebanker_loan_books`, `ecl_models`, `industry_types_updated`, `single_records`) that no migration knows; they are left alone and listed here.

Old test rows. The production copy held 213 loan-book rows with serial contract ids (1, 2, 3 ...) and sample balances in July 2024 to March 2025, from an early test import. The build retires them only when asked (`--retire-stale`), and only rows on a GL it covers or on no GL; the Mega Farm rows are kept.

The take-on population is 78, not 109. E-Banker's manual opening postings (operation date 31 July 2024: the 301 principal, the 303 opening interest, the 305 opening recovery, back-dated to each loan) exist for 78 accounts. The workbook's 109 facilities include ones closed before migration. The schedule balance at 31 July 2024 differs from the net opening principal on 72 of 73 mapped accounts, on both sides, because the schedules are contractual while the opening recovery is actual.

The revenue roll-forward reads the stage. Stage 3 interest is on the net basis, so the chain had to stage before the revenue run; the specification's order (EIR first, for discounting) still holds between EIR and ECL.

The feed's dates. E-Banker exports a timestamp column (LOS requests) with its time; the date gate now keeps the date part. One extract (security details) returns one row per security and account, so the register names a composite key and the gate checks uniqueness within the file.

5. Decisions and inputs MAIIC owes, found or confirmed by the build

Who	What	Why it matters now
Dr Thom or the reviewer of accounting rules	Approve the fee rulebook (30 rules seeded, three for E-Banker's own charge names)	19 of 31 solvable EIRs are blocked at the fee gate until a reviewer approves the rules and the classifications
Tamanda	Which of blocks 34 and 55 belongs to account 000104450000036; the fees of the take-on loans in the yellow columns or on the Take-on Schedules screen	The refused block; all 78 take-on loans start at the take-on balance until fees exist
Finance	The keyed GL openings of 1050201 and 1050202 (section 3.5)	Two golden checks pass only as accepted exceptions
Dr Thom	How the November 2025 Mega Farm book (3,490 loans, K54.85 bn) relates to the K51.5 bn gross and K48.7 bn Stage 3 of the statements; the monthly Mega Farm books	Section 16; the ECL golden number at 31 December 2025
Barry	September 2026 loan-book run (M09_03), the Mega Farm extracts (MF_01 to MF_08), the P2_13 re-run	The feed's first monthly pack; the Mega Farm build

6. What is still owed

Item	Spec	What it is	Size
Routes 2 to 5 of the feed	6.5	The scheduled export, the API push, the direct read, the vendor view: the same pack through the same door	Route 2: a script and a scheduler job; route 4 needs ICT's consent
Stress testing and the reports in the chain	6.10.1 steps 11, 12	Run from their screens today	Half a day each
FLI Adjustments screen and the fit approval	14.6	The chain's applied fits are proposals; a reviewer approves one on a screen and the route applies it per scenario	One day
The chain per scenario	15.5	The sensitivity uses the scenario multipliers; the full method runs the fitted relationship under each scenario's shocked path	One day after the fit approval
Macro screen, IMF WEO, RBM file	13	The five-tab screen, the WEO parser (the forecast years), the policy-rate file tab	One to two days
Audit workbooks: register, trace, pack; three modules	12 (CA-3, rest of CA-4)	The register where MAIIC signs each row, the per-record trace, the auditor's pack; IFRS 7 / IAS 1, the RBM directive, Contract Schedule 1	Two to three days
Mega Farm module	16	The programme in the ECL module under D30 with the three PD options	After Dr Thom's confirmation and the monthly books
Dark-mode sweep; help texts; the user manual's navigation chapter	11.6, 11.9	The pages that style elements directly; fifteen passages; screenshots	Ongoing; half a day for the texts
Tests for the PD and LGD services	6.10.1	The engines are proven by the bootstrap run; unit tests on a fixture are owed	Half a day
Golden numbers for the ECL and the revenue shift	6.10.2	Born on the first approved run	After the decisions in section 5

7. How to reproduce

```

mysql -uroot -e "CREATE DATABASE maiic_ifrs9_bootstrap" php artisan --env=bootstrap eir:bootstrap
--fresh --force-wipe --with-client-inputs --build --run-engines --verify php artisan --env=bootstrap eir:as-at
2025-12-31 php artisan --env=bootstrap fli:method-cards php artisan --env=bootstrap fli:correlate 2026-08
php artisan --env=bootstrap scenario:sets 2026-08 python tools/compliance/build_audit.py --all
--env=bootstrap php artisan test tests/Feature/Eir tests/Feature/Ebanker
tests/Feature/FLI/TransmissionMethodCatalogueTest.php tests/Feature/NavigationTest.php `

```

.env.bootstrap names the throwaway database and is git-ignored, like .env.demo. The test run above: 301 passed, 24 skipped. The 92 legacy medical tests and the five legacy FLI page tests fail on master before this work and are unrelated.

8. Files

Code: app/Services/Ebanker/{PackLandingService,LandingZoneReader,LoanBookBuildService,TakeonLandingService,ContractInputsBuildService}.php, app/Services/Eir/{StagingService,EirAsAtService}.php, app/Services/Macro/{WorldBankFetcherService,MacroImportService}.php, app/Services/Fli/TransmissionMethodCatalogue.php, app/Support/Fli/*, app/Console/Commands/{LandEbankerPack,BuildLoanBooks,LockReportingPeriod,StageLoanBooks,LandTakeonWorkbook,Bootstrap,EirAsAt,ImportWorldBankMacro,FliMethodCards}.php, app/Http/Controllers/{EirFeedController,EirTakeonController,EirAsAtController}.php, the migrations of 8 October 2026, config/menu.php, resources/js/{Layouts/AppLayout.vue,Jetstream/SidebarNav.vue,Jetstream/DropdownMenu.vue,navAccents.js,composables/useTheme.js,Pages/Eir/{Feed,Takeon,AsAt}/Index.vue}.

Inputs: docs/bootstrap/ (84 files in the manifest, including the original take-on workbook as received and the World Bank snapshot). Reference: docs/reference/ (the suite's engines and, from today, their support classes). Manuals: docs/manuals/installation/09-data-loading.md section 9.8 rewritten.